

Executive Summary

Project Objective

This project investigates what content, creator, and strategy factors drive ad efficiency (ROAS) and sales performance in TikTok's beauty and personal care category. We approach this question at two levels: first, an overall market trend analysis across 2,000+ beauty brands and 2,600+ ad videos; second, a deep-dive case study on Medicube, the #1 beauty seller on TikTok Shop, examining its boosting strategy, creator partnerships, and product portfolio. Our goal is to provide data-backed, actionable recommendations for beauty brands entering or scaling on TikTok and TikTok Shop.

Data & Analytical Rigor

We combined two complementary data sources: TikTok's Research API (3,184 beauty videos with engagement metrics and creator profile data) and Fastmoss, a third-party TikTok analytics platform (2,664 ad videos with ROAS and ad spend, plus 2,222 brand-level records). Because neither dataset alone could answer our research questions. TikTok API lacks ad performance data while Fastmoss lacks creator follower information (such as their follower amount, which is an important metric we use to divide creators into different tiers). We merged the two on creator username and video ID, enabling cross-dataset analyses such as linking creator tier to ROAS.

Our methods were chosen to match each question's structure. For the market-level analysis, we used multiple linear regression to test content features as ROAS predictors, TF-IDF with K-Means clustering and ANOVA to examine hashtag theme effects on engagement and ROAS, K-Means clustering on brand-level metrics to identify strategy archetypes, and linear regression to assess ad duration's effect on revenue across brand clusters.

For the Medicube case study, we employed Propensity Score Matching (PSM) to estimate the causal effect of video boosting on sales — deliberately excluding view count as a matching covariate since it lies on the causal pathway (Boost → Views → Sales). We validated this through three model specifications and confirmed robustness via Standardized Mean Difference diagnostics. We also ran log-log OLS regression with HC3 robust standard errors on 6,000 videos, ANOVA with Tukey HSD post-hoc tests on creator tier GPM (gross merchandise

per 1,000 views), and rank-based methods (Spearman correlation, Kruskal-Wallis) for the 32-product portfolio analysis where the small sample size precluded parametric approaches.

Results

Overall Market Trends: Surface-level content features (video duration, like rate, comment rate, hashtag count) do not predict ROAS (all $p > 0.05$, $R^2 \approx 0.001$). Hashtag topic clustering reveals significant engagement differences across content themes (ANOVA $F = 24.04$, $p < 0.001$) but no significant ROAS differences ($p = 0.65$) — high engagement does not translate to high ad returns. Brand strategy clustering shows that ROAS remains remarkably stable (4.2–4.5) across all four brand size clusters, from small brands spending ~\$27K to mega brands spending ~\$14.6M. Ad duration significantly predicts revenue only for small brands (+\$451/day, $p < 0.001$, $R^2 = 8.6\%$), not for larger brands.

Medicube Case Study: Boosting has a large causal effect on sales, with approximately \$9,200 in incremental 90-day revenue per video (95% CI: \$7,900–\$10,800), representing a ~20× lift over matched organic videos. The mechanism is primarily view amplification: views are the dominant sales driver ($\beta = 0.74$, 4× larger than any other factor), and our regression model explains 87% of sales variance. Even after controlling for views, boosted videos earn 121% more, suggesting boosting also improves audience targeting. For creator selection, mega creators (>1M followers) deliver 56% higher mean GPM (\$59.76) than other tiers (ANOVA $p < 0.001$), but with high internal variance ($\sigma = \$83$); nano, micro, and macro tiers perform statistically equivalently. On the product side, price is the strongest barrier to sales on TikTok Shop (Spearman $\rho = -0.51$, $p = 0.003$ for price vs. volume). The low-price tier (\$18–35) generates 65% of total revenue, and a single hero SKU, with the PDRN Pink Collagen Balm at \$24.50, accounts for 52% of all Medicube revenue. Time on market does not predict sales velocity; winning products declare themselves within 3–4 weeks.

Technical Report

Alignment with Course Rubric

Python Syntax & Data Collection

We used Python extensively throughout the project. Data collection was performed via TikTok's Research API and Fastmoss. For TikTok API, writing custom Python scripts was required to query JSON responses into structured DataFrames. For Fastmoss, we only need to export the data from the website. The API was called through VS Code running locally (details in Section 2) and the resulting CSV files were uploaded to Google Drive for collaborative analysis in Google Colab.

Data Analysis with Pandas and Matplotlib

All data cleaning, feature engineering, and statistical analysis were conducted in Python using Pandas, NumPy, and Statsmodels. Visualization was done with Matplotlib and Seaborn. Specific analyses included: Multiple Linear Regression, TF-IDF Vectorization, K-Means Clustering, Propensity Score Matching, One-way ANOVA, Tukey HSD Post-Hoc Test, Spearman Rank Correlation, and Kruskal-Wallis Test.

Identifying Relevant and Accessible Data Sources

We identified two complementary data sources: TikTok's Research API for engagement and creator profile data, and Fastmoss (a third-party TikTok analytics platform) for ad performance metrics including ROAS and ad spend. A key analytical contribution was designing the merge strategy between these datasets using creator username as the join key, which enabled cross-dataset analyses that neither source could support independently.

Translating Managerial Questions into Data Plans

We began with a broad managerial question: "What drives ad efficiency on TikTok Shop?" and decomposed it into four testable hypotheses for the overall market analysis: (1) Do content features predict ROAS? (Multiple regression, null hypothesis: all coefficients = 0.) (2) Do different hashtag themes produce different ROAS? (ANOVA, H_0 : group means are equal.) (3) Do brands of different scales achieve different ROAS? (K-Means + descriptive comparison.) (4) Does ad campaign duration predict revenue? (Linear regression stratified by brand cluster.) Each hypothesis was paired with a specific statistical method and a clear decision rule ($p < 0.05$).

Building on the overall market findings, we then narrowed our focus to a single brand - Mediube, one of the top-performing Korean beauty brands on TikTok shop and decomposed our investigation into four causal and exploratory hypotheses: (5) Does video boosting causally increase sales? (Propensity Score Matching, $H_0: ATT = 0$.) (6) Among views, followers, duration, boosting status, and creator category, which factors drive 90-day store revenue, and by how much? (Multiple OLS regression with standardized β , H_0 : all coefficients = 0.) (7) Do influencer tiers (Nano / Micro / Macro / Mega) differ in conversion efficiency measured by GPM? (One-way ANOVA + Tukey HSD post-hoc, H_0 : tier means are equal.) (8) Does product price correlate with sales performance, and how concentrated is Medicube's revenue across SKUs? (Spearman rank correlation + Kruskal-Wallis on price tertiles, H_0 : $\rho = 0$ / group medians are equal.) Each hypothesis was matched to a method appropriate for its data structure and sample size, with consistent decision rules ($p < 0.05$) and robustness checks across all four sections.

2. Technologies Used

VS Code (local)	Running TikTok Research API scripts locally, as Google Colab's environment could not execute the API cli
Google Colab	Primary analysis environment. All data cleaning, merging, feature engineering, statistical testing, and visual
Pandas / NumPy	Data manipulation, cleaning (filtering low-quality records, deduplication, type conversion), feature engineering
Matplotlib / Seaborn	All data visualizations including bar charts, boxplots, scatter plots, and regression line overlays for presentation
Scikit-learn	TF-IDF vectorization (TfidfVectorizer) and K-Means clustering for both hashtag topic clustering and brand strategy

Statsmodels / SciPy	OLS regression with significance testing, ANOVA (scipy.stats.f_oneway), and correlation analysis for all hypothesis
Gemini	Assisted with generating presentation visuals and helped organize and structure data within Colab notebook
Claude	Used to optimize statistical analysis strategy, specifically to determine which tests were appropriate for experiments
Fastmoss	Third-party TikTok analytics platform used as a secondary data source for ad-level metrics (ROAS, ad spend)

3. Challenges Encountered

Challenge 1: TikTok Shop API Access Denied

Our original plan was to use TikTok Shop's official API, which would have provided direct access to ad performance data, cart attachment status, and conversion metrics. However, we discovered that TikTok Shop API access requires merchant-level credentials that we could not obtain as academic researchers. This forced us to split our data pipeline into two parts: collecting engagement data through TikTok's Research API (which we could access), and supplementing it with third-party ad performance data from Fastmoss. While this workaround was effective, it introduced the dataset merging challenge described below.

Challenge 2: Bot Detection Blocked Comment Scraping

We originally intended to scrape video comments to perform sentiment analysis, a feature that would have been a significant differentiator for the project, as comment sentiment could serve as a proxy for purchase intent or brand perception. However, TikTok's anti-bot detection system flagged our scraping attempts and blocked access. We also attempted to determine whether individual videos had shopping cart links attached (a key indicator of commerce intent), but this data was not available through any accessible endpoint. The absence of these variables limited our ability to analyze the qualitative content dimension of ad performance.

Challenge 3: Cross-Dataset Integration

The most significant analytical challenge was figuring out how to combine TikTok API data with Fastmoss data. The two datasets had different time scopes, different units of observation, and no obvious shared key. We debated whether to join on product category, brand name, or creator identity. Ultimately, we chose to merge on creator username, which allowed us to link TikTok API's follower-based creator tier classification to Fastmoss's ROAS data.

Challenge 4: TikTok API Pagination and Data Volume

TikTok's Research API operates by surfacing videos through a feed-refresh mechanism, returning a maximum of 500–600 videos per query session. To build a dataset large enough for meaningful statistical analysis, we ran 5 separate collection sessions across different beauty-related hashtags (#beauty, #skincare, #makeup, etc.) and then deduplicated by video ID. This process yielded 3,184 unique videos after cleaning. The implication is that our sample represents a snapshot of TikTok's beauty content rather than a comprehensive census. The API's feed-based architecture means we captured what the algorithm surfaced during our collection windows, which may skew toward higher-engagement content. We acknowledge this as a potential selection bias in our analysis.